Probability & Statistics 1

1			Q1: if consistent "0.8" incorrect or $^1/_8$, $^7/_8$ or 0.02 allow M marks in ii , iii & 1st M1 in i
i	Binomial stated $0.9437 - 0.7969$ or $^8C_3\times0.2^3\times0.8^5$ $= 0.147$ (3 sfs)	M1 M1 A1 3	or implied by use of tables or $^8C_3$ or $0.2^a\times0.8^b$ $(a+b = 8)$
ii	$1 - 0.7969$ $= 0.203$ (3 sf)	M1 A1 2	allow $1 - 0.9437$ or 0.056(3) or equiv using formula
iii	8×0.2 oe 1.6	M1 A1 2	$8 \times 0.2 = 2$ M1A0 $1.6 \div 8$ or $^1/_{1.6}$ M0A0
Total		**7**	
2	first two d's $= \pm1$ Σd^2 attempted (= 2) $1 - \dfrac{6 \times \text{"2"}}{7(7^2-1)}$ $= {}^{27}/_{28}$ or 0.964 (3 sfs)	B1 M1 M1dep A1	S_{xx} or $S_{yy} = 28$ B1 $S_{xy} = 27$ B1 $S_{xy}/\sqrt{(S_{xx}S_{yy})}$ M1 dep B1 1234567 & 1276543 (ans $^2/_7$): MR, lose A1
Total		**4**	
3 i	x independent or controlled or changed Value of y was measured for each x x not dependent	B1 1	Allow Water affects yield, or yield is dependent or yield not control water supply Not just y is dependent Not x goes up in equal intervals Not x is fixed
ii	(line given by) minimum sum of squs	B1 B1 2	B1 for "minimum" or "least squares" with inadequate or no explanation
iii	$S_{xx} = 17.5$ or 2.92 $S_{yy} = 41.3$ or 6.89 $S_{xy} = 25$ or 4.17 $r = \dfrac{S_{xy}}{\sqrt{(S_{xx}S_{yy})}}$ $= 0.930$ (3 sf)	 B1 M1 A1 3	or $91 - 21^2/_6$ or $394 - 46^2/_6$ B1 for any one or $186 - {}^{21\times46}/_6$ dep B1 0.929 or 0.93 with or without wking B1M1A0 **SC** incorrect n: max B1M1A0
iv	Near 1 or lg, high, strong, good corr'n or relnship oe Close to st line or line good fit	B1ft B1 2	$\|r\|$ small: allow little (or no) corr'n oe Not line accurate. Not fits trend
Total		**8**	